

Vayu: India Real-Time Airfare Price Index

Methodology note (supporting document, SIH 2026, ByteBrigade). Describes the prototype as implemented in the submitted repository.

1. What the index measures

Vayu is a daily, fixed-basket price index for domestic air travel between 26 major Indian airports (all 325 unordered pairs). It is built like a CPI elementary index: a fixed basket of routes, fixed basket weights, fixed reference prices, and only the current price varying. The purpose is a timely fare-movement signal to complement the periodic airfare sub-indices in official statistics, not to replace them.

2. Pipeline

Stage	What it does	Key parameters (from code)
Collect	Headless browser (Scraping / Chromium) loads the Google Flights results for each route and captures the fares returned.	Travel date = run date + 21 days (one advance-purchase window). 4 s delay between requests, retries with exponential backoff.
Normalise	Groups raw fares by route and reduces each route to one representative price.	Median of all fares seen for the route that day. Fares outside Rs 1,000 to Rs 1,00,000 are rejected at parse time.
Filter	Drops observations that look like parsing errors rather than real fare moves.	(a) Day-over-day: change beyond +/-150% versus last known good price. (b) Cross-sectional: robust MAD z-score of price per km above 4.0, applied only when 5 or more routes were observed in the run.
Index	Computes the weighted index from the routes that passed the filter today.	Formula in section 3; coverage reported on every result.
Publish	Upserts one row per date to Supabase; dashboard reads it. Each route in the table is tagged 'Live scraped data' or 'Sample / last known' with a coloured dot.	Runs daily at 03:17 UTC (08:47 IST), full 325-route basket; on-demand quick run covers the top 20 routes by weight.

3. Index formula

$$\text{index}(t) = 100 \times \frac{\sum_i (w_i \times P_i(t) / P_i(\text{base}))}{\sum_i (w_i)}$$

The sums run only over routes i that were scraped and passed the filter on date t . w_i is the fixed basket weight and $P_i(\text{base})$ the fixed reference price of route i . Because the covered routes' weights are divided by their own total, the weights renormalise to 1.0 on each run. The index therefore never uses stale or placeholder prices; routes without fresh data appear in the route table (marked 'Sample / last known') but contribute nothing to the headline number. The share of total basket weight that was actually covered is published next to the index (coverage_weight_pct).

4. Basket weights

Weights follow a gravity model: weight is proportional to $(\text{mass}_A \times \text{mass}_B) / \max(\text{distance}_{AB}, 600 \text{ km})$, normalised so all 325 weights sum to 1.0. The 600 km floor stops short hops that road and rail absorb from outranking trunk routes. The top routes by weight are BOM-BLR, BOM-HYD, DEL-BOM, BLR-HYD and BLR-MAA. The airport traffic-mass column is an illustrative relative scale (Delhi = 100), **not official DGCA data**. Replacing it with DGCA sector-wise passenger traffic and re-running one script regenerates every downstream weight.

5. Reference prices and how to read the level

The reference price $P_i(\text{base})$ is currently a modelled fare schedule, not a surveyed base period: $(\text{Rs } 2,200 + \text{Rs } 3.6 \times \text{route distance in km}) \times \text{a deterministic route-specific factor between } 0.96 \text{ and } 1.04$. The dashboard's base date is the date of the first pipeline run. **Consequently the index level (about 152 at the time of writing) is not a measure of inflation since the base date.** It says the median scraped fare is about 1.5 times the modelled schedule. What is meaningful today is the direction and size of day-to-day change, on comparable coverage.

Re-basing the reference prices to a surveyed base period (so that the base day reads exactly 100) is the first roadmap item.

6. Data sources and what is verified

Source	Status
Google Flights	Only source in the scheduled workflow. Provides the published index.
IndiGo fare-radar, Air India air-bounds	Response parsers written against real captured responses (test fixtures dated 2026-09-27). Not part of the scheduled index run.
MakeMyTrip	Tested and blocked by bot defences; excluded.
SpiceJet, Akasa	Not implemented (configuration placeholders only).

Automated tests: 95 unit tests across parsing, normalisation, anomaly filtering, index computation and page actions, all passing when run against the submitted repository. These test code behaviour on fixtures; they are not a validation of the index against official statistics.

7. Known limitations

Limitation	Effect	Planned fix
Placeholder reference prices (section 5)	Index level not interpretable as inflation	Survey a real base period; set it to 100
Placeholder traffic masses (section 4)	Route weights are illustrative	Use DGCA sector-wise traffic
Single source, single advance window	No cross-source or window comparison	Enable verified airline endpoints; add further windows
Coverage varies run to run	Small day-on-day moves can reflect a different route mix	Publish coverage beside the index (done); add a matched-route change measure
No explicit cabin filter beyond the site's default view	Fare mix may include other cabins	Filter to economy fare families at parse time
Median of all fares seen for a route	Includes flights at all times of day	Weight by departure-time bands or seat capacity